SMR Group 9 — Presentation Script

Speaker 1: Leroy (Slides 1–5)

Slide 1 — Introduction

Good afternoon, everyone. Today we're looking at small modular nuclear reactors, or SMRs, and asking a practical sustainability question: where could they actually reduce greenhouse gas emissions, and what might prevent that from happening?

We've chosen a real design and a specific materials improvement.

I'll introduce the technology and our proposal. Sam will explain the lifecycle and carbon calculations, and Victor will assess the Australian barriers and our final recommendation.

Slide 2 — Coal is closing. What replaces its firm power?

As coal stations retire, electricity systems must replace their reliable supply when wind and solar output are low. That is where SMRs might help.

This slide compares two emissions figures. The New South Wales grid's published operational electricity factor is about 640 grams of carbon dioxide equivalent per kilowatt-hour.

A United Nations study estimates around five to six grams per kilowatt-hour across the full lifecycle of conventional nuclear electricity.

Importantly, these are different accounting boundaries: one is operational grid emissions, and the other includes nuclear construction, fuel and end-of-life. They are not a direct like-for-like comparison.

They do, however, show why replacing fossil-fuel generation with low-carbon nuclear power might offer a climate benefit.

But can an SMR arrive in time at a reasonable cost?

We use these figures to motivate the question, not as proof that any particular SMR project will cut that much carbon.

Slide 3 — The BWRX-300

Our case study is GE Hitachi's BWRX-300, a boiling-water reactor designed to generate about 300 megawatts of electricity.

The idea is to repeat smaller, standardised units with more factory manufacturing.

The design also uses natural water circulation and isolation condensers for passive removal of decay heat after shutdown. Some cooling functions therefore work without powered pumps, although risk is not zero.

The manufacturer also claims more than a fifty percent reduction in concrete use per megawatt compared with its larger ESBWR design.

We treat that as a vendor claim, not a verified environmental result.

Even a smaller reactor still contains substantial quantities of concrete and steel, which creates an opportunity for materials engineering.

Slide 4 — Darlington case study

We selected Darlington in Ontario because it gives us a real project rather than a purely hypothetical design.

Ontario Power Generation applied for a licence in October 2022, received it in April 2025, and secured provincial approval for Unit 1 in May.

The programme budget is about twenty point nine billion Canadian dollars for four planned reactors, including shared infrastructure and preparatory work.

The first unit is expected to enter service around 2030, although that remains a project schedule, not an operating result.

This provides a real setting for our materials proposal.

Slide 5 — Our proposed materials initiative

Our proposal is to investigate using low-clinker limestone-calcined clay cement, often called L C three, in suitable concrete works for future Darlington units two to four.

Cement production emits carbon dioxide through kiln heating and clinker chemistry. LC three replaces part of that clinker with calcined clay and limestone.

We would begin in non-safety-related works and only consider reactor structures after testing and qualification.

It is not an approved nuclear substitution.

For our screening calculation we use an estimate of fifty thousand cubic metres of concrete for one BWRX-300 from a proposed project in Poland.

That is not a measured Darlington quantity.

Our Australian comparison at Liddell is a separate scenario, not the location of this material trial.

This keeps the materials proposal measurable: we can compare one concrete mix directly against another.

Sam will now explain how we assessed the potential savings.

Speaker 2: Sam (Slides 6–9)

Slide 6 — Lifecycle approach and material inventory

Our sustainability assessment goes beyond power-station emissions.

Our overall lifecycle boundary covers uranium mining and milling, conversion and enrichment, fuel fabrication, construction, operation, and finally decommissioning and waste management.

Our comparison uses one megawatt-hour delivered to the grid.

Our own calculation covers construction materials, not a measured whole-reactor lifecycle.

We use Granta EduPack as a framework for construction material analysis, and published lifecycle studies for nuclear fuel and waste.

Our preliminary bill of materials comes from the Polish BWRX-300 project estimate: fifty thousand cubic metres of concrete, six thousand tonnes of structural steel and eight thousand tonnes of modular steel.

We avoid double-counting steel components.

We then compare the concrete baseline with a potential lower-carbon alternative.

Material mass, processing routes and transport distances all affect the final embodied-emissions estimate.

Slide 7 — Published concrete emissions and calculated savings

Here is the most important quantitative part of our proposal.

We use Pillai and colleagues' published comparison of ordinary Portland concrete and LC three in the same M50 strength category.

Their cradle-to-gate assessment reports three hundred and eighty kilograms of carbon dioxide equivalent per cubic metre for the conventional mix, versus two hundred and seventy kilograms for the LC three mix.

That is a reduction of one hundred and ten kilograms per cubic metre, or about twenty-nine percent.

The reported twenty-eight-day strengths were sixty and fifty-six point seven megapascals respectively.

That does not establish nuclear suitability.

For one reactor we apply the Polish fifty-thousand-cubic-metre estimate, assuming forty percent is eligible.

Forty percent gives twenty thousand cubic metres.

Multiplying that by one hundred and ten kilograms gives two million two hundred thousand kilograms, or two thousand two hundred tonnes of carbon dioxide equivalent potentially avoided per unit.

This is a screening scenario, not a measured Darlington saving. The eligible share needs validation.

Slide 8 — Technical feasibility and sensitivity

Emissions are only part of the engineering decision.

First, the fresh concrete has to be workable. Around steel plates, it must fill tight spaces without voids. We would test slump flow, filling ability and early strength.

Second, hydration heat, shrinkage and Ontario's freeze-thaw conditions could affect durability.

We would therefore test heat evolution, permeability, cracking and long-term durability.

Third, any use in safety-related nuclear structures would require the relevant design standards, quality assurance and regulatory acceptance.

Our safest starting point is non-safety-related concrete.

Our sensitivity results are eleven hundred, twenty-two hundred and thirty-three hundred tonnes saved at twenty, forty and sixty percent eligibility.

These are alternative assumptions, not measured eligibility rates. They show exactly which uncertainty drives our result.

A lower-emissions material that fails qualification cannot deliver any real project saving.

Slide 9 — Comparison of annual avoided emissions

Next, how much grid carbon might the reactor avoid?

Location matters because the result depends on the generation it replaces.

At three hundred megawatts and an assumed ninety percent capacity factor, the reactor would generate approximately two point three seven terawatt-hours per year.

Using today's average grid emissions figures, the slide estimates around one point five million tonnes of carbon dioxide equivalent avoided annually in New South Wales, compared with roughly seventy thousand to one hundred and sixty thousand tonnes in Ontario.

Ontario's grid is already lower carbon than NSW's.

These are not forecasts: a plant displaces particular generators, not necessarily the grid average.

And an Australian SMR, if built years from now, would enter a changing grid.

Timing and displaced generation can outweigh the concrete saving.

Victor will now consider the barriers to deploying the technology.

Speaker 3: Victor (Slides 10–14)

Slide 10 — Legal barriers in Australia

The first barrier is legislation.

Our concrete proposal concerns Darlington, where there is an established nuclear regulator and a construction programme.

Liddell in New South Wales is a very different situation.

Australia's federal Environment Protection and Biodiversity Conservation Act restricts approval of nuclear power plants, the federal radiation-protection legislation limits the national regulator's licensing powers, and New South Wales has its own prohibition on nuclear power reactors.

Those laws would have to change before an SMR at Liddell.

Australia would also need regulators, skilled workers and consultation.

In comparison, Darlington's construction licence process took about twenty-nine months in a country with an existing nuclear industry.

Legal permission is therefore only the beginning of the delivery challenge.

Slide 11 — Cost barriers

The second barrier is economics.

SMRs promise repeat manufacturing, but first projects face major risks.

In Australia, CSIRO's GenCost analysis identifies nuclear power as a high-cost new-build option compared with renewable generation supported by storage and other firming.

Darlington's twenty point nine billion Canadian dollar budget covers four units and shared infrastructure, not a single reactor.

Another cautionary example is NuScale's proposed project in Idaho.

Its target electricity price increased from fifty-eight to eighty-nine US dollars per megawatt-hour in reported project estimates, and the project was cancelled in 2023.

Those prices use different base years.

Our LC three proposal might lower concrete-related emissions, but it cannot by itself eliminate financing, manufacturing or project delivery risks.

In fact, any material that delays construction could make project economics worse.

Slide 12 — Social licence and safety

The third barrier is public and community acceptance.

In the 2024 Lowy Institute survey, sixty-one percent of Australians supported nuclear electricity and thirty-seven percent opposed it.

But a national opinion poll does not tell us whether people living near Liddell would consent to a reactor.

The risk comparison on this slide also reminds us that coal's air pollution causes health harms, while nuclear concerns focus on accidents and waste.

Both matter.

A credible project would need an independently supervised safety case, a funded waste-management pathway, transparent engagement with local and First Nations communities, and clear local benefits.

Social sustainability means that people affected by the project have a meaningful say in it.

Slide 13 — Fuel security and geopolitical concentration

Our fourth barrier is the nuclear fuel supply chain.

Australia has major uranium resources, but possessing uranium does not mean being able to manufacture finished reactor fuel.

Fuel production also requires conversion, enrichment and fabrication, often involving facilities in other countries.

Our sources estimate Australia's recoverable uranium share at twenty-eight percent and Russia's enrichment capacity at around forty-three percent.

We also use the Herfindahl-Hirschman Index, or H H I, to describe how concentrated that market is.

It sums the squares of suppliers' market shares, and our preliminary estimate for enrichment is around three thousand, indicating substantial concentration.

Because this estimate is preliminary, it requires confirmation of source-year shares.

For an SMR programme, supply security might be improved through diversified contracts and trusted fuel suppliers.

But uranium reserves alone do not guarantee energy sovereignty.

Slide 14 — Triple bottom line and conclusion

To conclude, we separate our two findings.

First, at Darlington, LC three concrete could reduce construction emissions.

Using published concrete data and a clearly stated forty percent eligibility assumption, we estimate a potential saving of twenty-two hundred tonnes of carbon dioxide equivalent per reactor.

That is worthwhile only if performance is proven without delaying delivery.

Second, the Australian question is broader.

SMRs offer low-carbon electricity, but their climate value depends on actual displaced generation, project costs and when they begin operating.

Environmentally, nuclear is low carbon, but the net benefit depends on deployment.

Economically, capital cost and delivery risk remain major weaknesses.

Socially, safety, waste, regulation and local consent must be addressed.

We recommend staged LC three qualification at Darlington, starting in non-safety works, while treating an Australian SMR as a separate uncertain scenario.

Thank you. We welcome your questions.

Slide 15 — References

Leave the references slide displayed while answering questions. The references do not need to be read aloud.